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Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am applying for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) on the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 with Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours. My thesis used solid-state NMR with diffraction to relate molecular motion in organic and metal-organic crystals to their macroscopic behaviour.

The chapter closest to this project concerns a solid solution. I synthesised UiO-68-type zirconium MOFs containing a terphenyl rotor linker and a longer azobenzene photoswitch linker in nominal ratios from 10 to 70%. Because the linkers differ in length, the PXRD reflections, which are dominated by the Zr_6 clusters, shifted to lower angle as the photoswitch content increased, and periodic DFT models (CASTEP) with ordered linkers predicted extra reflections that were absent. Together these pointed to a random distribution of linkers within one phase. ^{13}C CP-MAS confirmed that both linkers were incorporated. With a deuterated rotor linker I recorded ^{2}H spectra between 178 and 338 K, fitted with 90° and 180° jumps above 10^7 Hz using electric field gradients calculated in CASTEP. After loading hexafluorobenzene, which I confirmed by ^{19}F MAS and quantified by ^{19}F NMR with standard additions, the 180° jumps were suppressed and the 90° site populations depended on the photoswitch state (ACS Mater. Lett., 2026). In molecular solid solutions whose components have similar size and shape, diffraction will distinguish far less, and solid-state NMR becomes the main probe of how the components are distributed and how they move.

In a toluene solvate of a pyrrolo[3,2-b]pyrrole that jumps when cooled to 77 K (Mater. Chem. Front., 2026), I prepared the toluene-d_8 analogue, recorded ^{2}H spectra from 173 to 338 K and simulated the lineshapes in NMR WebLab with quadrupolar parameters calculated in CASTEP (GIPAW). The fit required a three-site methyl rotation and 180° ring flips, both above 10^7 Hz at 248 K. Variable-temperature SCXRD gave a thermal expansion coefficient of 271 MK^{-1} oblique to the face in contact with the cold surface, and we attributed the jump to this anisotropic expansion. This work was done with Dr Martin Dračínský and Dr Jan Blahut at IOCB Prague. I also acquired the ^{13}C CP-MAS and ^{2}H data for cocrystals whose channels hold acetone and tetrahydrofuran in different ratios, where the salient temperature follows the solvent composition (Angew. Chem. Int. Ed., 2026).

I ran these experiments on a Bruker 500 MHz spectrometer (^{1}H/X MAS probe), and also acquired ^{31}P and ^{15}N CP-MAS spectra and analysed ^{13}C T_1 and solid-state ^{1}H and ^{19}F relaxometry data. I made the compounds in my thesis, including the deuterated analogues, and characterised them by SCXRD, PXRD, DSC and TGA. I have not yet used DNP. The UK DNP MAS NMR Facility in Manchester, and the chance to extend this work to ^{17}O and natural-abundance ^{15}N, are among my main reasons for applying.

I have 14 publications. During my PhD I co-supervised four bachelor's theses and nine research placements and managed purchasing and inventory for our laboratory. I currently work as an Industrial Property Expert at the Mexican Institute of Industrial Property; my contract ends in December 2026 and I could start in January 2027. Thank you for considering my application.

Yours sincerely,

Dazaet Galicia-Badillo, PhD